BOTANY PRACTICALS

I - Preparation and Demonstration of Slides

Note: Teacher has to prepare a temporary slide using fresh specimen for demonstration. (During examination permanent slides can be used if temporary slide preparation is not possible)

Exercise 1: T.S. of Mature anther

Aim: To study and identify the given slide – T.S of Anther

Principle: Androecium is made up of stamens. Each stamen possesses an anther and a filament. Anther bears pollen grains which represent the male gametophyte.

Requirements: Anther of *Datura metel*, glycerine, safranin, slide, cover slip, blade, brush, needle to prepare temporary slides, permanent slide of T.S. of mature anther and compound microscope.



Collect buds and opened flowers of *Datura metel*. Dissect the stamens, separate the anthers and take thin sections and observe the structure under the microscope. Record the various stages of anther from your observation.

Diagnostic Features

- A mature anther is bilobed (dithecous) and the two lobes are joined by a connective.
- Each anther lobe has two pollen chambers in which pollen grains are produced.
- A microsporangium or pollen sac is surrounded by four wall layers. They are epidermis , endothecium, middle layers and tapetum.
- Centre of the microsporangium (pollen sac) is filled with haploid pollen grains.

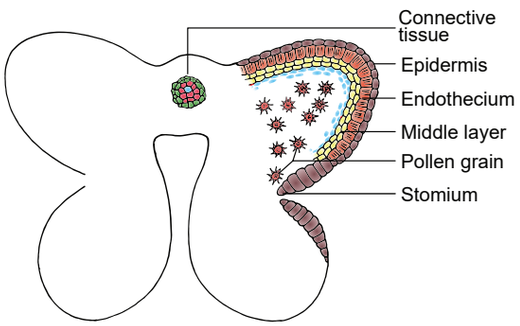


Figure 1: Pollen grain stage of anther

Exercise 2: L.S of an Angiospermic ovule.

Aim: To study and identify the L.S. of an Angiospermic Ovule.

Principle: In female reproductive part of a flower, the basal swollen part is ovary. The ovules are present inside the ovary, later they develops to seed.

Requirement: Permanent slide of L.S. of Ovule, microscope

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